

Fashion Inventory Analysis

By Pringgondani Ade

PROJECT OVERVIEW

Retail businesses often face challenges in managing inventory efficiently. Excess inventory ties up capital in unsold products and generates markdown losses when items must be discounted to clear stock. This project analyzes fashion inventory data to identify patterns contributing to **deadstock, markdown losses, and capital tied up** — examining product attributes such as size, color, and price tier to uncover factors affecting sales performance and inventory risk.

DEADSTOCK RATE	MARKDOWN LOSS	CAPITAL TIED UP	TOTAL LOSS
25.75%	\$241,867	\$366,507	\$608,374
Of total inventory unsold	Revenue lost to discounts	Locked in unsold stock	Combined financial impact

PROBLEM STATEMENT

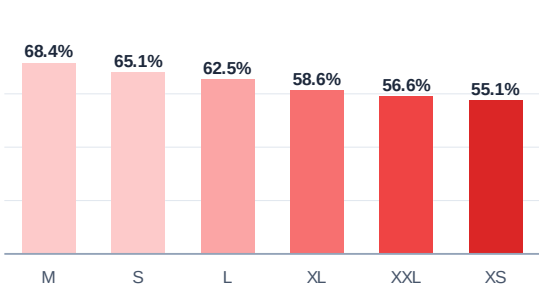
This fashion business carries a **25.75% deadstock rate** resulting in **\$241K markdown loss** and **\$366K capital tied up** — a combined **\$608K total financial exposure** from inefficient inventory planning.

RESEARCH QUESTIONS

- What product characteristics are most associated with deadstock items?
- What cost savings could be achieved through optimized inventory decisions?
- What recommendations can reduce the deadstock rate?

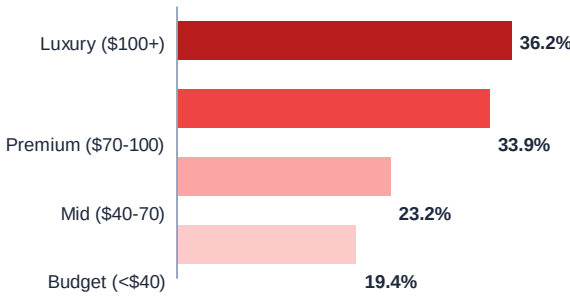
KEY FINDINGS

Average Sell-Through by Size



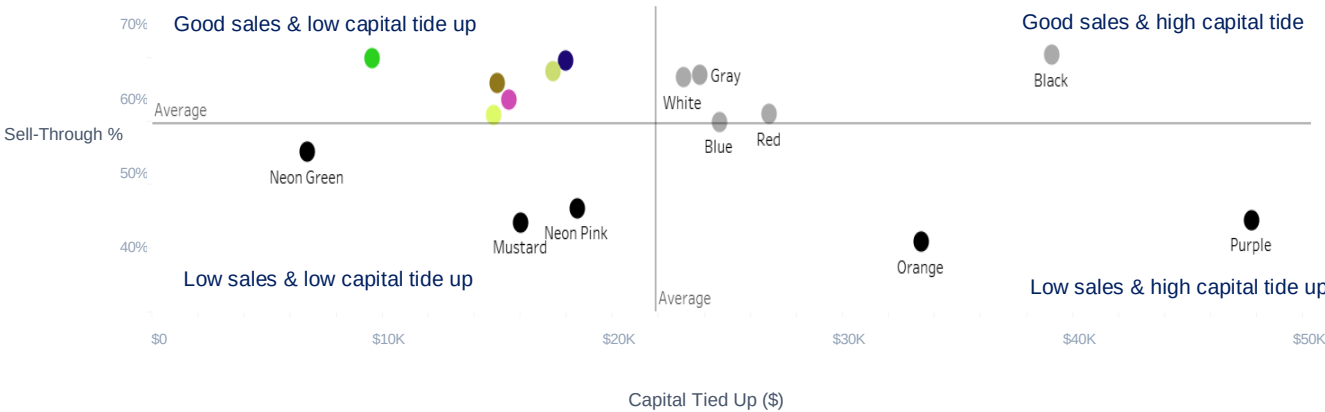
Size **M** leads at **68.4%** sell-through, while extreme sizes **XS** (55.1%) and **XXL** (56.6%) underperform by **12–13%** indicating slower turnover and higher deadstock risk.

Deadstock Rate by Price Tier



Luxury (\$100+) and Premium (\$70–100) carry the highest deadstock rates at 36.2% and 33.9% — **70% of all deadstock** concentrated in these two high-price segments.

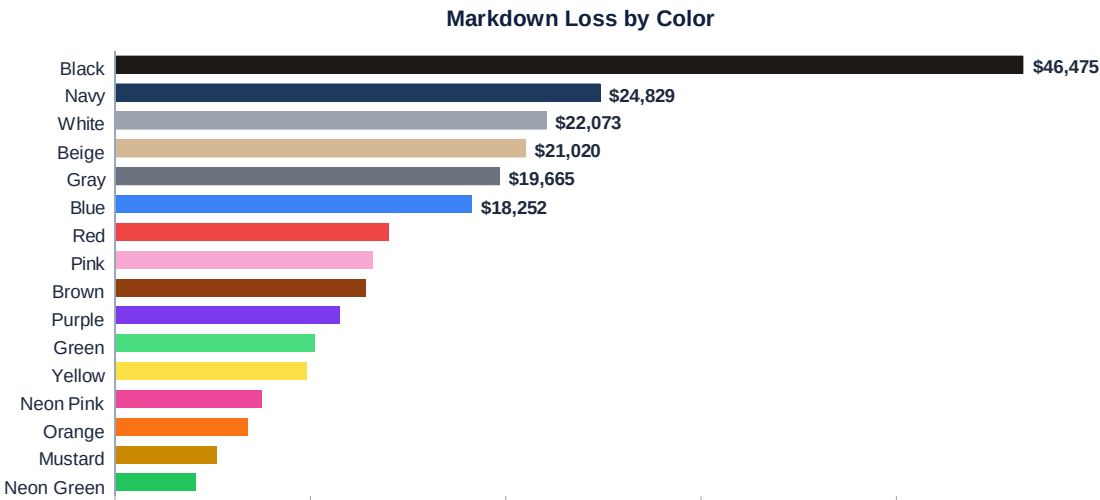
Capital Tied Up vs. Average Sell-Through by Color



Bold colors (Neon Green, Mustard, Neon Pink, Orange, Purple) show **below-average sell-through and low capital tied up** — low demand across the board. **Neutral colors** (Black, White, Gray, Blue, Red) generate higher sales but carry **dangerously high capital tied up**, indicating significant overstocking.

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Black alone accounts for \$46,475 in markdown loss. The top 5 neutral colors (Black, Navy, White, Beige, Gray) collectively drive \$134,062 — 55% of total markdown loss, confirming that overstocking in popular neutral colors is the primary driver of markdown exposure.

INSIGHTS & RECOMMENDATIONS

01 Optimize Size Allocation

Extreme sizes (XS and XXL) underperform by 12–13% compared to mid-range sizes, indicating slower inventory turnover and higher deadstock risk. Reducing order quantities for these sizes by 20–30% could improve inventory efficiency and unlock up to 10–15% in additional sell-through value. Optimize inventory allocation by prioritizing high-performing sizes and reducing extreme sizes.

02 Reduce High-Tier Price Inventory

Higher-priced items (Luxury & Premium) have significantly higher deadstock rates, exceeding mid-range products by over 10–16%. Optimizing inventory allocation by reducing high-tier stock by 20–30% could lower deadstock exposure and free up approximately 10–15% of tied capital.

03 Rebalance Neutral Color Orders

Black, White, Gray, Beige, and Navy show 25–35% excess inventory above optimal levels. Black alone has \$78K in tied-up capital. Reduce neutral-color order quantities by 20% and implement demand-driven replenishment based on historical sell-through rates. This Optimization could unlock ~\$250K in capital; significantly reduce markdown risk on neutral colors.

04 Deprioritize Bold & Neon Colors

Bold colors (Neon Green, Mustard, Neon Pink, Orange, Purple) show consistently below-average sell-through below the 50% mark. Consider limiting these to small-batch trial orders only (10–20% of current volume) and monitor response.

Bottom Line: The \$608K total loss is not a demand problem — it is an **inventory planning problem**. The data consistently points to three fixable root causes: wrong size mix, over-investment in high-price tiers, and overstocking in neutral colors. Correcting these three levers through demand-driven replenishment and right-sizing could recover an estimated **\$300K+ in capital and markdown savings**.